

MH1100 Calculus I

Handout 0 - Course Overview

100 Prisoners Problem

The director of a prison offers 100 death row prisoners, who are numbered from 1 to 100, a last chance. A room contains a cupboard with 100 drawers. The director randomly puts one prisoner's number in each closed drawer. The prisoners enter the room, one after another. Each prisoner may open and look into 50 drawers in any order. The drawers are closed again afterwards. If, during this search, every prisoner finds their number in one of the drawers, all prisoners are pardoned. If even one prisoner does not find their number, all prisoners die. Before the first prisoner enters the room, the prisoners may discuss strategy — but may not communicate once the first prisoner enters to look in the drawers. What is the prisoners' best strategy?

Surprisingly, there is a strategy that provides a survival probability of more than 30%.

Financial Derivatives: Options

Imagine that Apple's stock is currently trading at \$180 per share. You happen to own 100 shares of this stock. However, your assessment suggests that it is unlikely the share price will exceed \$190 within the next month. In light of this, you approach James and extend an offer: the option to purchase your shares on or before September 15th, at a fixed price of \$190 per share. This arrangement prompts a few key considerations:

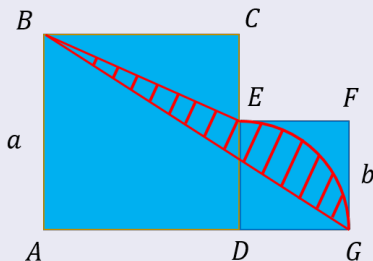
- Do you need to provide a premium to James to solidify this trade agreement?
- Does James need to offer you a premium instead?
- What is the appropriate premium amount for either party?

To delve into the mechanics of premium calculation, a grasp of Stochastic Calculus is essential.

Welcome to MH1100—Calculus I

Welcome Question

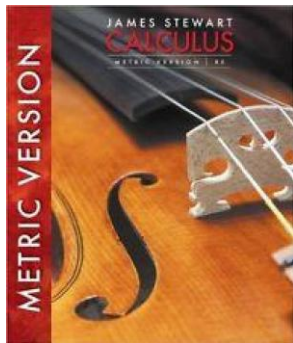
$ABCD$ and $DEFG$ are two squares with side lengths a and b . Find the area of the shaded region.



Comment: Before Calculus was developed, problems of finding areas, volumes, and lengths of curves were so difficult that only a genius could meet the challenge. But now, these challenging problems are accessible to all of us after learning Calculus.

Course Information

- Textbook: Calculus, *James Stewart*, The 8th, metric edition, Cengage Learning.



- I recommend getting a copy of this because: (1) This course will be closely based on it; (2) It will also (probably) be used for Calculus II & III.

Course Information

- Lecturers: TONG Ping (Weeks 1-7, tongping@ntu.edu.sg),
XIA Kelin (Weeks 8-13, xiakelin@ntu.edu.sg)
- Time:
 - Lectures (Weeks 1-7, 8-13):
[Thursdays 9:30 am - 10:20 am](#) LT 27,
[Fridays 9:30 am - 11:20 am](#) LT 23.
 - Tutorials (Weeks 2-7, 8-13)
[Tuesday 09:30-10:20, 11:30-12:20, 13:30-14:20](#) at SPMS-TR+3 or
SPMS-TR+5 or SPMS-TR+6
[Wednesday 11:30-12:20, 13:30-14:20, 16:30-17:20](#) at SPMS-TR+3 or
SPMS-TR+4 or SPMS-TR+5 or SPMS-TR+7.
- Office hours
 - Weeks 2-8: [Tuesday 03:30 pm-4:30 pm, Thursday 03:30 pm-4:30 pm.](#)
[Venue: Tong Ping's Office SPMS-MAS-04-17.](#)
 - Weeks 9-13: [Professor Xia will provide this information in Week 8.](#)

- Tutorials start from Week 2
- Every week we will upload a sheet of about 10 problems on Friday for the next week tutorial.
- We will highlight a few of these (maybe 3 or 4) which your tutor should definitely discuss in the tutorial. We will write out solutions to all the problems and upload solutions after the tutorial (Wednesday evening).
- The remaining part of the tutorial is up to your tutor.

About MH1100: Calculus I

A lot of what we will discuss you have actually seen before.

So what are the goals?

- We will aim for a deeper and more precise understanding of calculus
 - Calculus is the mathematical study of continuous change.
 - *needed for proving* → **Definitions** will be extremely important. Learn them precisely!
 - Learning to carefully **explain your deductions logically**.
 - A taste of **proofs**.
- Our basic aim: to build **a strong foundation**. A lot of mathematics depends on calculus, so we must proceed slowly but precisely.
- Prepare for the future: (1) A fundamental course in your undergraduate program; (2) Calculus is used in every branch of the physical, chemical, and computer sciences, business, medicine and in other fields wherever a problem can be mathematically explained.

By the end of Week 7, you are supposed to be able to solve the following 4 questions.

Example 1

Consider the function

$$f(x) = \begin{cases} 1, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$

Case 1: If $|x| \leq 1$, $f(x) = 1$
 $\therefore f[f(x)] = f(1) = 1$
 Case 2: If $|x| > 1$, $f(x) = 0$
 $f[f(x)] = f(0) = 1$
 $\therefore f[f(x)] = 1$ for all cases
 $\therefore f\{f[f(x)]\} = f(1) = 1$
 $\therefore f\{f[f(x)]\} = 1$ for all values of x

Find $f\{f[f(x)]\}$.

Example 2

Use the ϵ - δ definition to prove that $\lim_{x \rightarrow 3} (x^2 - 3x + 1) = 0$.

Example 3

Let $f(x) = x \ln x$, find $f^{(n)}(1)$ ($n \geq 2$).

Example 4

The derivative function $f'(x)$ of a differentiable function $f(x)$ is not necessary to be continuous.

Consider the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

- (a) Show that $f(x)$ is continuous in its domain.
- (b) Show that $f(x)$ is differentiable everywhere.
- (c) Show that $f'(x)$ is not a continuous function.

Some comments

- The lectures are a bit like tutorials. We try a lot of problems during the lectures to help you understand the concepts.
- Every one or two weeks, we will load a printable version of the Lecture Slides named Handout $\#$ ($\# = 0, 1, \dots$) on the website before the lectures. These detailed handouts can serve as the lecture notes as well.
- Please let me know when you do not follow what I am saying.

Where to get help

5 ways to get help:

- In tutorials
- Meet the lecturer during the [Office Hours](#)
- Send me an email at tongping@ntu.edu.sg
- Ask your tutor for help
- From the [Math Clinic](#).

Course Evaluation

- Homework assignment I on 15 September: Lectures up to 15 September, Tutorials 1–4; Return to TONG Ping by 21 September (6 pm);
- Midterm Test on 29 September: Lectures up to 28 September, Tutorials 1–6;
- Homework assignment II on 03 November: Lectures up to 03 November, Tutorials 1–10; Return to XIA Kelin by 09 November (6 pm);
- Final Exam ($\approx$ Chapters 1–3): Lectures up to week 13, Tutorials 1–12, 2 hours, no help sheet allowed.

Components of evaluation:

- 10% Homework I
- 20% Midterm Test
- 10% Homework II
- 60% Final exam